

This document reports the results of a corpus audit (v2, updated 2026-05-08) conducted against the indexed knowledge base (272 peer-reviewed papers) to assess whether each proposed evaluation question can be fairly answered by the RAG chatbot. Q1 and Q3 have been revised since the first audit; a source DOI was also supplied for Q4. For each question we checked whether the expected answer values appear verbatim in the parsed text or table chunks, and identified any structural reasons the chatbot may succeed or fail.

Q1 Highest methane yield from anaerobic co-digestion with algal biomass (corpus-wide synthesis)

OK GOOD — KE

Question type: Corpus-wide synthesis — find the maximum reported value across 272 papers
Source paper: Not pinned to a single DOI; answer depends on what the corpus contains
RAG suitability: Appropriate — hybrid mode retrieves top-k chunks on methane yield; LLM synthesises

This is a well-designed synthesis question. The corpus contains 272 papers on algae-AD integration, many reporting methane yields for co-digestion experiments. The RAG chatbot will retrieve the most semantically relevant chunks and attempt to identify the highest reported value along with the associated co-substrate, species, and operating conditions.

Caveats:

- RAG retrieves by semantic similarity, not by magnitude. If the paper with the absolute highest yield is under-represented in the top-k retrieved chunks, the chatbot may report a high but not the highest value. This is an inherent ceiling for any top-k RAG system.
- GPT without retrieval would likely give a plausible but generic or stale answer (training data cut-off, no source citation). The RAG chatbot should outperform GPT here on citation accuracy and specificity — making this an excellent discriminating question.
- To make scoring unambiguous, identify in advance (from literature review) the paper and value you consider the correct answer, so you can check whether the chatbot's citation matches the ground-truth source.

Q2 Cultivation mode for *C. reinhardtii* in Table 2 — DOI: 10.1016/j.biotechadv.2025.108581

OK GOOD — KE

Paper in corpus: Yes — indexed as selecting_optimal_algal_strains.pdf
Expected answer: Batch culture
Table 2 parsed: Yes — full HTML table structure captured including 'Cultivation mode' column

Table 2 was successfully parsed by MinerU into a structured HTML table with columns for Algal species, Stress, Cultivation mode, Evolutionary duration, Change in phenotype, and Reference. The *Chlamydomonas reinhardtii* row should be present; the chatbot needs to locate the correct row and return 'Batch culture'. This is an appropriate test of table-specific retrieval and a question GPT is likely to hallucinate or answer generically.

Caveat:

- Verify that the *C. reinhardtii* row appears in the portion of the table that was captured in the chunk (large tables are sometimes split). A quick manual check of the paper is recommended before finalising this question.

Q3 Operational stage effects on CO₂ removal in HRAP system — DOI: 10.1016/j.biortech.2023.129955

++ STRONG — KE

Paper in corpus: Yes — indexed as 1-s2.0-S0960852423013834-main.pdf
Expected: 91.1% CO₂-REOUND verbatim — stage VI text: 'CO₂-RE = 91.1%' with 3.4% residual CO₂

Expected: 94.2% CO₂-REBOUND verbatim — stage VII text: 'CO₂-RE = 94.2%' with 2.2% residual CO₂
ammonium stripping / pH-REBOUND — mentioned in body text describing stage transitions
LNP (nanoparticle) addition — referenced in stage VII context

All key expected values (91.1%, 94.2%, 3.4% CO₂) appear verbatim in text chunks of this paper, not in figures. The operational stage descriptions (pH adjustment, ammonium stripping, biogas recirculation, liquid nanoparticle addition) are all present in the indexed text. This is now one of the two strongest questions in the set. It requires the chatbot to synthesise a multi-stage narrative from a single paper — something GPT is likely to either mis-state or report without a specific citation.

Note on scoring:

- The expected answer covers multiple stages. Accept the response if it correctly identifies the 91.1% and 94.2% CO₂ removal efficiency values, attributes them to pH adjustment and LNP addition respectively, and notes the biomass-shading decline at the end of stage VII. Citation to DOI 10.1016/j.biortech.2023.129955 is required for full credit.

Q4 Biomethane purity & HRAP-AD vs PSA economic trade-off — DOI: 10.1016/j.cej.2022.138323

~ PARTIAL

98.9% CH₄ purity: FOUND — verbatim in insights_microalgaebased_technologies.pdf, parsed table
Source paper for PSA figures 10.1016/j.cej.2022.138323 — NOT indexed as a standalone file
83.3% OpEx reduction vs PSA NOT FOUND — source paper absent from corpus (only in ref lists)
123.5% CapEx increase vs PSA NOT FOUND — same reason

The 98.9% CH₄ purity figure is confirmed in the indexed corpus and will be retrievable. However, the DOI provided for the PSA comparison (10.1016/j.cej.2022.138323) corresponds to a paper that is cited by other indexed papers but is not itself in the knowledge base. The 83.3% and 123.5% figures do not appear anywhere in the 272-paper index. The chatbot may give a reasonable qualitative answer about HRAP vs PSA trade-offs from related papers, but cannot retrieve these specific quantitative values.

Required action to promote to 'Keep':

- Add DOI 10.1016/j.cej.2022.138323 (Chemical Engineering Journal, 2022) to the papers folder and re-run ingest.py for that file only. Once indexed, Q4 becomes an excellent two-part test: single-paper lookup (98.9% CH₄ purity) plus cross-paper comparison (HRAP vs PSA costs from the CEJ paper).

Q5 Best methane improvement from algae + wheat straw co-digestion (synthesis question)

++ STRONG — K

Expected answer: 77% higher methane production vs microalgae mono-digestion
Found in chunks: Yes — verbatim in body text of 1-s2.0-S0960852417304339-main.pdf

The exact finding — 'The methane yield increased by 77% with the co-digestion as compared to microalgae mono-digestion' — appears verbatim in a parsed list-content chunk of an indexed paper. This is the strongest question in the set. It tests whether the chatbot can retrieve a specific quantitative result that GPT would likely approximate, misattribute, or state with lower confidence. Recommend keeping this question unchanged.

Summary

Q	Status	Verdict	Recommended Action
Q1	Corpus-wide synthesis; methane yield data p	Keep	Pre-register expected answer + source DOI for scoring
Q2	Table 2 parsed; 'Cultivation mode' column pr	Keep	Verify C. reinhardtii row is in captured chunk
Q3	91.1%, 94.2%, 3.4% CO2 found verbatim in t	Keep	Strong multi-stage question; all values confirmed
Q4	98.9% CH4 confirmed; PSA source paper no	Partial	Index DOI 10.1016/j.cej.2022.138323 to complete Q4
Q5	77% value found verbatim in indexed body te	Keep	No changes needed — strongest single-lookup question

General Recommendations

- The set is now substantially stronger after the Q1 and Q3 revisions. Q1 (corpus-wide synthesis), Q3 (multi-stage HRAP text), and Q5 (verbatim co-digestion yield) are all ready to use. Q2 needs a quick manual check; Q4 needs one additional paper indexed.
- Index DOI 10.1016/j.cej.2022.138323 to unlock Q4. This requires downloading the PDF, placing it in the papers/ folder, and running: `python ingest.py` (it will skip already-indexed files and process only the new one).
- For each question, record the source paper DOI alongside the expected answer before running the evaluation. This makes scoring unambiguous and lets you verify whether the chatbot's in-text citation matches the ground-truth source.
- Run a parallel baseline: ask GPT-4o (no retrieval) the same five questions. Q2 tests table lookup (GPT likely gives a generic cultivation mode), Q3 tests multi-stage quantitative recall (GPT unlikely to cite 91.1% or 94.2% correctly), and Q5 tests specific yield data. These are the three clearest cases where RAG provides a verifiable factual advantage.